

Evaluating the Bias-Variance Trade-off

In this project, we'll explore the relationship between the APSE across a variety of samples using a fictional dataset of hotel front desk information, called `hotel_frontdesk.csv`.

It has the following variables.

`hotel_frontdesk.csv` has the check-ins and check-outs. Each row is either a check-in or a check-out, and it has:

Variable	Description
<code>guest_id</code>	Unique id for every guest
<code>in_timestamp</code>	The YYYY-MM-DD hh:mm:ss of the time of check in
<code>out_timestamp</code>	The YYYY-MM-DD hh:mm:ss of the time of check out
<code>length_of_stay</code>	Integer number of nights stayed 1-7
<code>room_id</code>	Room number XXYY, X: Floor, Y: Room
<code>floor</code>	Floor number, where ground = 1
<code>room_on_floor</code>	Room on floor.
<code>days_booked_ago</code>	Days before check in that the hotel reservation was made
<code>price</code>	Price paid for the total stay, 2018 Canadian Dollars
<code>week</code>	week of the summer, 1-13
<code>in_ts_hour</code>	integer hour from check in timestamp 00-23
<code>out_ts_hour</code>	integer hour from check out timestamp 00-23
<code>in_day_of_week</code>	check in sun, mon, ... , sat
<code>out_day_of_week</code>	check out sun, mon, ... , sat

We can also get `price_per_night`, which will be our response variable y , with the following code.

```
dat_hotel = read.csv("data/hotel_frontdesk.csv")
dat_hotel$price_per_night = dat_hotel$price / dat_hotel$length_of_stay
```

- a) (1 mark) Open the dataset `dat_hotel` and use the `by()` command to get $\tau(x)$, the mean of `price_per_night` for each week. Call you answer, the output of `by()`, `tau`. Report it, and use it and the following code to append the values to your dataset.

This will append `tau` to `dat_hotel`

```
tau = by(dat_hotel$price_per_night, dat_hotel$week, mean)
tau
```

```
## dat_hotel$week: 1
## [1] 343.8239
## -----
## dat_hotel$week: 2
## [1] 166.467
## -----
```

```
## dat_hotel$week: 3
## [1] 673.3062
## -----
## dat_hotel$week: 4
## [1] 172.6002
## -----
## dat_hotel$week: 5
## [1] 383.7976
## -----
## dat_hotel$week: 6
## [1] 407.605
## -----
## dat_hotel$week: 7
## [1] 285.8566
## -----
## dat_hotel$week: 8
## [1] 530.7397
## -----
## dat_hotel$week: 9
## [1] 225.8629
## -----
## dat_hotel$week: 10
## [1] 469.6022
## -----
## dat_hotel$week: 11
## [1] 226.0649
## -----
## dat_hotel$week: 12
## [1] 446.8593
## -----
## dat_hotel$week: 13
## [1] 176.584
## -----
## dat_hotel$week: 14
## [1] 96.25235
## -----
## dat_hotel$week: 15
## [1] 70.30814
```

```
taunum = as.numeric(tau)
dat_hotel$tau = taunum[dat_hotel$week]
```

- b) (1 mark) Find the irreducible error of this data. That is, find the $\text{Ave}(\text{Var}(Y|x))$. The variable `tau` that you appended to the dataset in part a will be helpful.

```
mean((dat_hotel$price_per_night - dat_hotel$tau)^2)
```

```
## [1] 29506.55
```

- c) (2 marks) Using `predict()` and `lm()`, find the APSE for a polynomial model of degree 3 of `price_per_night` as a function of `week`.

```
Y = dat_hotel$price_per_night
X = dat_hotel$week
model = lm(Y ~ X + I(X^2) + I(X^3), data = dat_hotel)

summary(model)
```

```
##
## Call:
## lm(formula = Y ~ X + I(X^2) + I(X^3), data = dat_hotel)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -349.55 -161.92  -49.12   109.78  1039.78
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  375.15470    25.85585   14.509  < 2e-16 ***
## X              5.81964    13.83466    0.421  0.67405
## I(X^2)         1.90173     2.03799    0.933  0.35086
## I(X^3)        -0.24026     0.08719   -2.755  0.00591 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 227.1 on 2091 degrees of freedom
## Multiple R-squared:  0.1019, Adjusted R-squared:  0.1006
## F-statistic: 79.06 on 3 and 2091 DF,  p-value: < 2.2e-16
```

```
yhat = predict(model)
apse = mean((Y - yhat)^2)
apse
```

```
## [1] 51481.75
```

- d) (2 marks) Fill out the following table, and comment on the trend and the minimum value, and why you should expect to see these trends.

```
apse_vec = rep(0, 15)

for (d in 0:14) {

  if (d == 0) {
    fit = lm(Y ~ 1, data = dat_hotel)
  } else {
    fit = lm(Y ~ poly(week, d, raw = TRUE), data = dat_hotel)
  }

  preds = predict(fit)
  apse_vec[d+1] = mean((Y - preds)^2)
}

apse_vec
```

```
## [1] 57320.96 54600.07 51668.69 51481.75 50184.69 49267.06 49228.46 49112.09
## [9] 44888.45 41827.55 40716.61 36646.78 36223.89 36223.89 34039.26
```

Polynomial Degree	APSE($\mathcal{P}, \mu_{\mathcal{P}}$)
0	57320.96
1	54600.07
2	51668.69
3	51481.75
4	50184.69
5	49267.06
6	49228.46
7	49112.09
8	44888.45
9	41827.55
10	40716.61
11	36646.78
12	36223.89
13	36223.89
14	34039.26

What we can see here is that as the model gets more and more complex, APSE gets smaller since it is measuring on the same data that was used to fit the model, and the model has more flexibility to chase the data.

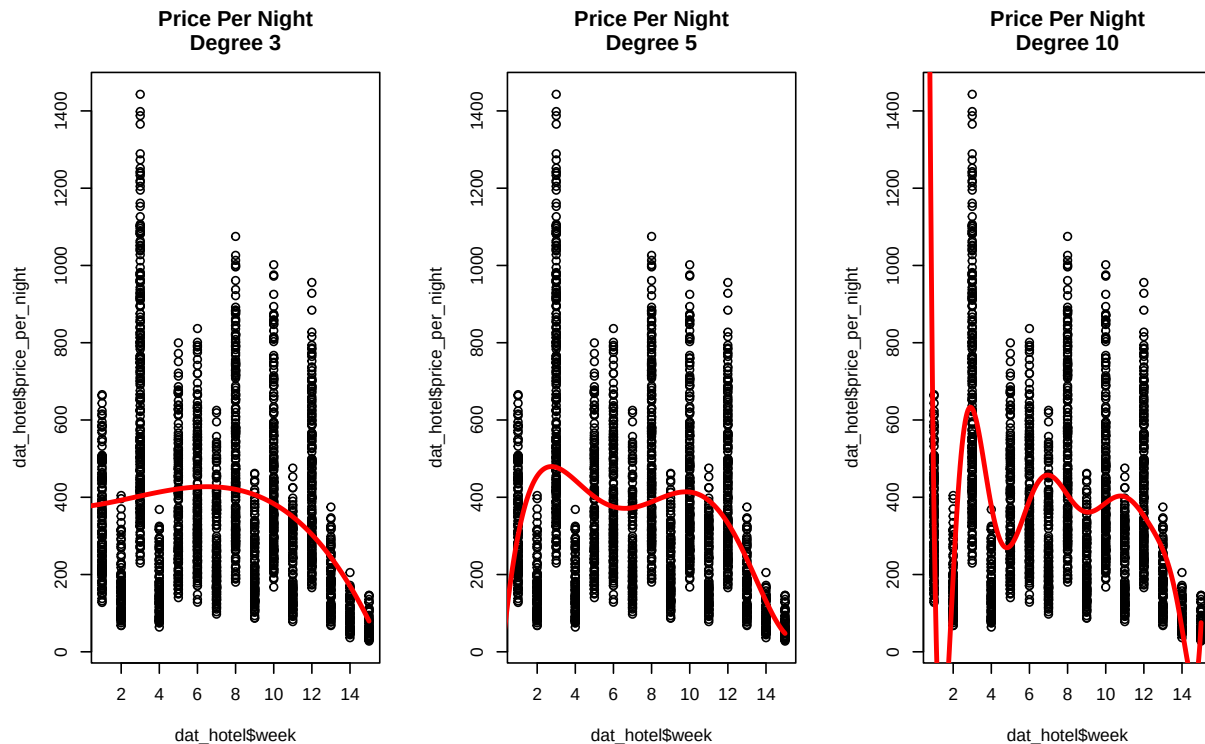
- e) (3 marks) Fill in the following code to plot the fitted values for polynomials of degree 3, 5, and 10. Comment on which of these models would you trust the most to fit the next 4 weeks of data, if we had it.

```
par(mfrow = c(1,3))
weekseq = seq(from=0, to=15, by=0.01)
newdat = data.frame(week = weekseq)

# First plot, degree 3
mod = lm(price_per_night ~ poly(week, 3, raw = TRUE), data=dat_hotel)
polyfit = predict(mod, new=newdat)
plot(dat_hotel$price_per_night ~ dat_hotel$week, main= "Price Per Night\nDegree 3")
lines(polyfit ~ weekseq, lwd=3, col="Red")

# Second plot, degree 5
mod = lm(price_per_night ~ poly(week, 5, raw = TRUE), data=dat_hotel)
polyfit = predict(mod, new=newdat)
plot(dat_hotel$price_per_night ~ dat_hotel$week, main= "Price Per Night\nDegree 5")
lines(polyfit ~ weekseq, lwd=3, col="Red")

# Third plot, degree 10
mod = lm(price_per_night ~ poly(week, 10, raw = TRUE), data=dat_hotel)
polyfit = predict(mod, new=newdat)
plot(dat_hotel$price_per_night ~ dat_hotel$week, main= "Price Per Night\nDegree 10")
lines(polyfit ~ weekseq, lwd=3, col="Red")
```



I would trust the degree-5 model the most as it captures enough trend in the data, in which the degree-3 model fails to do so and does not provide enough pattern in comparison, and the degree-5 model also does not oscillate crazy like the degree-10 model as it has too much flexibility.

- f) (3 marks) Use `subset` or something similar to split your data so that weeks 1-12 are your training set `dat_hotel_train`, and weeks 13-15 are your test set `dat_hotel_test`. Apply your polynomial models as you did in part d, but trained only on weeks 1-12. Fill out the table for the APSE for predicted values for weeks 13-15. **Notice that this table only goes to degree 10**

You may use scientific notation to fill the table.

```
dat_hotel_train = subset(dat_hotel, week <= 12)
dat_hotel_test  = subset(dat_hotel, week > 12)

Y_test = dat_hotel_test$price_per_night
X_test = dat_hotel_test$week

apse_test = rep(0, 11)

for (d in 0:10) {
  if (d == 0) {
    fit = lm(price_per_night ~ 1, data = dat_hotel_train)
  } else {
    fit = lm(price_per_night ~ poly(week, d, raw = TRUE), data = dat_hotel_train)
  }

  preds = predict(fit, newdata = data.frame(week = X_test))
  apse_test[d+1] = mean((Y_test - preds)^2)
```

```
}
apse_test
```

```
## [1] 8.058107e+04 7.446983e+04 5.871521e+04 2.423986e+05 3.059147e+04
## [6] 4.206111e+06 8.717343e+07 9.037261e+07 1.219604e+10 3.589258e+09
## [11] 7.065892e+12
```

Polynomial Degree	APSE($\mathcal{T}, \mu_S$)
0	8.058107e+04
1	7.446983e+04
2	5.871521e+04
3	2.423986e+05
4	3.059147e+04
5	4.206111e+06
6	8.717343e+07
7	9.037261e+07
8	1.219604e+10
9	3.589258e+09
10	7.065892e+12

g) (2 marks) Is the overall trend in part f what you would expect, why or why not?

Yes, this is exactly the trend we expect. In part (f), the APSE is evaluated on the test set rather than on the data used to fit the model. As model complexity increases, the predictor becomes more tightly tailored to the training sample, which lowers the bias but greatly increases the variance. This overfitting leads to poor out-of-sample prediction performance, so the test set APSE eventually increases rather than continuing to decrease, which is exactly what the bias–variance trade-off is.

h) (2 marks) Is the minimum APSE on or near the model you would expect it to be, why or why not?

The min APSE is on the model with degree 4, which is near what we expected in part (e), and this is still what we expected for the same reason: model with higher complexity tends to be overfitting and not capture the real trend while models with low complexity fail to capture enough trend so the one in the middle would be doing a better job.